Generative AI & LLMs

Lecture: LLM Evaluation – Metrics & Benchmarks

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Agenda

- **LLM Evaluation Metrics**
  - Accuracy
  - ROUGE (Recall-oriented Understudy for Gisting Evaluation)
  - BLEU (Bilingual Evaluation Under Study)
- **Evaluation Benchmarks**
  - GLUE
  - SuperGLUE
  - HELM

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LLM Evaluation Metrics

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LLM Evaluation - Challenges

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

"Mike really loves drinking tea." ❤️☕ = "Mike adores sipping tea." ❤️☕

"Mike does not drink coffee." 🤢☕ ≠ "Mike does drink coffee." 🧐☕

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LLM Evaluation - Metrics

ROUGE

- Used for text summarization
- Compares a summary to one or more reference summaries

BLEU SCORE

- Used for text translation
- Compares to human-generated translations

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Metrics

- **ROUGE** stands for **Recall-Oriented Understudy for Gisting Evaluation**. It is a set of metrics used in Large Language Models (LLMs) to evaluate the quality of text summarization or generation by comparing the model's output against human-written reference summaries, focusing on how much of the reference content is recalled.

- **BLEU** (**Bilingual Evaluation Understudy**) is a metric used in LLM evaluation to measure the similarity between machine-generated text and human-provided reference text. Ranging from 0 to 1, it evaluates precision - how many words/phrases in the generated text appear in the reference - making it popular for machine translation.

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LLM Evaluation - Metrics - Terminology

n-gram

The dog lay on the rug as I sipped a cup of tea.

bigram

unigram

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LLM Evaluation - Metrics - ROUGE-1

Reference (human):
It is cold outside.

Generated output:
It is very cold outside.

$$\text{ROUGE-1 Recall} = \frac{\text{unigram matches}}{\text{unigrams in reference}} = \frac{4}{4} = 1.0$$

$$\text{ROUGE-1 Precision:} = \frac{\text{unigram matches}}{\text{unigrams in output}} = \frac{4}{5} = 0.8$$

$$\text{ROUGE-1 F1:} = 2\,\frac{\text{precision x recall}}{\text{precision + recall}} = 2\,\frac{0.8}{1.8} = 0.89$$

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LLM Evaluation - Metrics - ROUGE-1

Reference (human):
It is cold outside.

Generated output:
It is not cold outside.

$$\text{ROUGE-1 Recall} = \frac{\text{unigram matches}}{\text{unigrams in reference}} = \frac{4}{4} = 1.0$$

$$\text{ROUGE-1 Precision:} = \frac{\text{unigram matches}}{\text{unigrams in output}} = \frac{4}{5} = 0.8$$

$$\text{ROUGE-1 F1:} = 2\,\frac{\text{precision x recall}}{\text{precision + recall}} = 2\,\frac{0.8}{1.8} = 0.89$$

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LLM Evaluation - Metrics - ROUGE-2

Reference (human):
It is cold outside.

It is | is cold | cold outside

Generated output:
It is very cold outside.

It is | is very | very cold | cold outside

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LLM Evaluation - Metrics - ROUGE-2

Reference (human):

It is cold outside.

It is | is cold | cold outside

Generated output:

It is very cold outside.

It is | is very | very cold | cold outside

$$\text{ROUGE-2 Recall:} = \frac{\text{bigram matches}}{\text{bigrams in reference}} = \frac{2}{3} = 0.67$$

$$\text{ROUGE-2 Precision:} = \frac{\text{bigram matches}}{\text{bigrams in output}} = \frac{2}{4} = 0.5$$

$$\text{ROUGE-2 F1:} = 2\,\frac{\text{precision x recall}}{\text{precision + recall}} = 2\,\frac{0.335}{1.17} = 0.57$$

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LLM Evaluation - Metrics - ROUGE-L

Reference (human):

It is cold outside.

Generated output:

It is very cold outside.

Longest common subsequence (LCS):

It is

cold outside

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LLM Evaluation - Metrics - ROUGE-L

Reference (human):

It is cold outside.

Generated output:

It is very cold outside.

LCS:
Longest common subsequence

$$\text{ROUGE-L Recall:} = \frac{\text{LCS(Gen, Ref)}}{\text{unigrams in reference}} = \frac{2}{4} = 0.5$$

$$\text{ROUGE-L Precision:} = \frac{\text{LCS(Gen, Ref)}}{\text{unigrams in output}} = \frac{2}{5} = 0.4$$

$$\text{ROUGE-L F1:} = 2\,\frac{\text{precision x recall}}{\text{precision + recall}} = 2\,\frac{0.2}{0.9} = 0.44$$

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LLM Evaluation - Metrics - ROUGE hacking

Reference (human):

It is cold outside.

Generated output:

cold cold cold cold

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LLM Evaluation - Metrics - ROUGE clipping

Reference (human):
It is cold outside.

Generated output:
cold cold cold cold

$$\text{ROUGE-1 Precision} = \frac{\text{unigram matches}}{\text{unigrams in output}} = \frac{4}{4} = 1.0$$ 🥶

$$\text{Modified precision} = \frac{\text{clip(unigram matches)}}{\text{unigrams in output}} = \frac{1}{4} = 0.25$$

Generated output:
outside cold it is

$$\text{Modified precision} = \frac{\text{clip(unigram matches)}}{\text{unigrams in output}} = \frac{4}{4} = 1.0$$ 🤔

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LLM Evaluation - Metrics

ROUGE

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BLEU SCORE

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LLM Evaluation - Metrics - BLEU

BLEU metric = Avg(precision across range of n-gram sizes)

Reference (human):

I am very happy to say that I am drinking a warm cup of tea.

Generated output:

I am very happy that I am drinking a cup of tea. - BLEU 0.495

I am very happy that I am drinking a warm cup of tea. - BLEU 0.730

I am very happy to say that I am drinking a warm tea. - BLEU 0.798

I am very happy to say that I am drinking a warm cup of tea. - BLEU 1.000

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LLM Evaluation - Metrics

ROUGE

- Used for text summarization
- Compares a summary to one or more reference summaries

BLEU SCORE

- Used for text translation
- Compares to human-generated translations

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Evaluation Benchmarks

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Evaluation benchmarks

GLUE

SuperGLUE

HELM

MMLU (Massive Multitask Language Understanding)

BIG-bench

GLUE - General Language Understanding Benchmark

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GLUE

GLUE

The tasks included in SuperGLUE benchmark:

Corpus	|Train|	|Test|	Task	Metrics	Domain
Single-Sentence Tasks					
CoLA	8.5k	**1k**	acceptability	Matthews corr.	misc.
SST-2	67k	1.8k	sentiment	acc.	movie reviews
Similarity and Paraphrase Tasks					
MRPC	3.7k	1.7k	paraphrase	acc./F1	news
STS-B	7k	1.4k	sentence similarity	Pearson/Spearman corr.	misc.
QQP	364k	**391k**	paraphrase	acc./F1	social QA questions
Inference Tasks					
MNLI	393k	**20k**	NLI	matched acc./mismatched acc.	misc.
QNLI	105k	5.4k	QA/NLI	acc.	Wikipedia
RTE	2.5k	3k	NLI	acc.	news, Wikipedia
WNLI	634	**146**	coreference/NLI	acc.	fiction books

Source: Wang et al. 2018, "GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding"

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SuperGLUE

SuperGLUE

The tasks included in SuperGLUE benchmark:

Corpus	|Train|	|Dev|	|Test|	Task	Metrics	Text Sources
BoolQ	9427	3270	3245	QA	acc.	Google queries, Wikipedia
CB	250	57	250	NLI	acc./F1	various
COPA	400	100	500	QA	acc.	blogs, photography encyclopedia
MultiRC	5100	953	1800	QA	$F1_a$/EM	various
ReCoRD	101k	10k	10k	QA	F1/EM	news (CNN, Daily Mail)
RTE	2500	278	300	NLI	acc.	news, Wikipedia
WiC	6000	638	1400	WSD	acc.	WordNet, VerbNet, Wiktionary
WSC	554	104	146	coref.	acc.	fiction books

Source: Wang et al. 2019, "SuperGLUE: A Stickier Benchmark for General-Purpose Language Understanding Systems"

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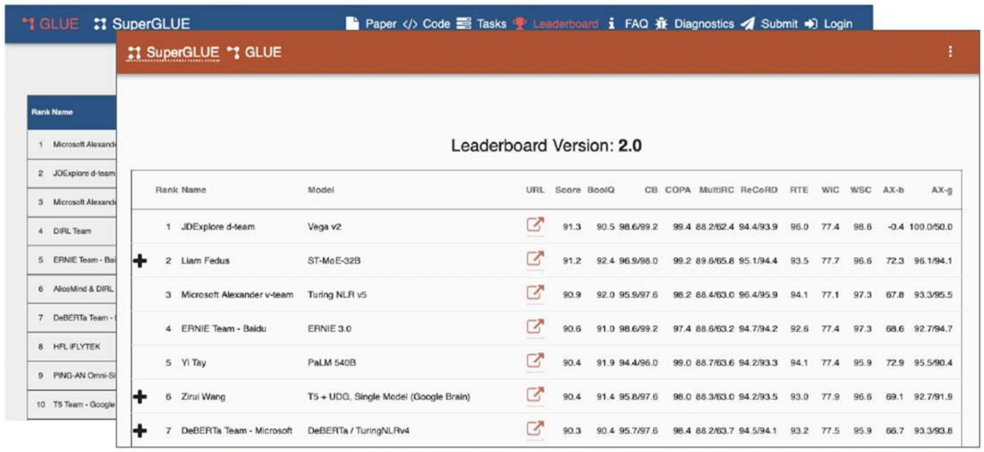
GLUE and SuperGLUE leaderboards

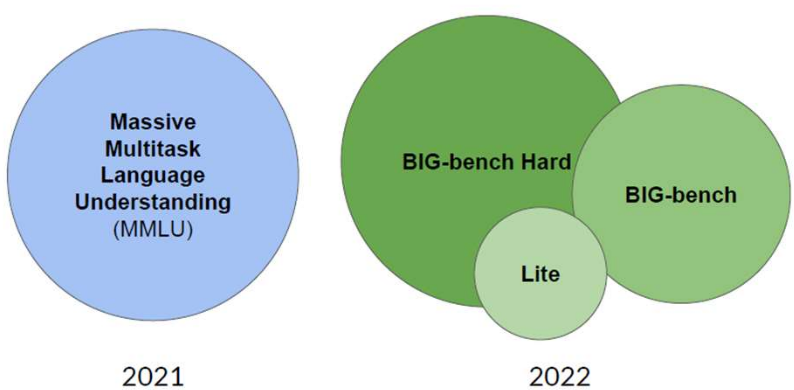


Leaderboard Version: **2.0**

	Rank	Name	Model	URL	Score	BoolQ	CB	COPA	MultiRC	ReCoRD	RTE	WiC	WSC	AX-b	AX-g
	1	JDExplore d-team	Vega v2		91.3	90.5	98.6/99.2	99.4	88.2/62.4	94.4/93.9	96.0	77.4	98.6	-0.4	100.0/50.0
+	2	Liam Fedus	ST-MoE-32B		91.2	92.4	96.9/98.0	99.2	89.6/65.8	95.1/94.4	93.5	77.7	96.6	72.3	96.1/94.1
	3	Microsoft Alexander v-team	Turing NLR v5		90.9	92.0	95.9/97.6	98.2	88.4/63.0	96.4/95.9	94.1	77.1	97.3	67.8	93.3/95.5
	4	ERNIE Team - Baidu	ERNIE 3.0		90.6	91.0	98.6/99.2	97.4	88.6/63.2	94.7/94.2	92.6	77.4	97.3	68.6	92.7/94.7
	5	Yi Tay	PaLM 540B		90.4	91.9	94.4/96.0	99.0	88.7/63.6	94.2/93.3	94.1	77.4	95.9	72.9	95.5/90.4
+	6	Zirui Wang	T5 + UDG, Single Model (Google Brain)		90.4	91.4	95.8/97.6	98.0	88.3/63.0	94.2/93.5	93.0	77.9	96.6	69.1	92.7/91.9
+	7	DeBERTa Team - Microsoft	DeBERTa / TuringNLRv4		90.3	90.4	95.7/97.6	98.4	88.2/63.7	94.5/94.1	93.2	77.5	95.9	66.7	93.3/93.8

Disclaimer: metrics may not be up-to-date. Check https://super.gluebenchmark.com and https://gluebenchmark.com/leaderboard for the latest.

Benchmarks for massive models



Source: Hendrycks, 2021. "Measuring Massive Multitask Language Understanding"

Source: Suzgun et al. 2022. "Challenging BIG-Bench tasks and whether chain-of-thought can solve them"

Holistic Evaluation of Language Models (HELM)

HELM

Metrics:
1. Accuracy
2. Calibration
3. Robustness
4. Fairness
5. Bias
6. Toxicity
7. Efficiency

Models

Scenarios

	J1-Jumbo	J1-Grande	J1-Large	Anthropic-LM	BLOOM	T0pp	Cohere XL	Cohere Large	Cohere Medium	Cohere Small	GPT-NeoX
NaturalQuestions (open)			✔	✔	✔	✔	✔	✔	✔	✔	✔
NaturalQuestions (closed)	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
BoolQ	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
NarrativeQA	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
QuAC	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
HellaSwag	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
OpenBookQA	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
TruthfulQA	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
MMLU	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
MS MARCO				✔	✔		✔	✔	✔	✔	✔
TREC				✔	✔		✔	✔	✔	✔	✔
XSUM	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
CNN/DM	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
IMDB	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
CivilComments	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
RAFT	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔

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Holistic Evaluation of Language Models (HELM)

Center for Research on Foundation Models HELM Models Scenarios Results Raw runs v0.2.2 (last updated 2023-03-19)

Core scenarios

The scenarios where we evaluate all the models.

[Accuracy | Calibration | Robustness | Fairness | Efficiency | General information | Bias | Toxicity | Summarization metrics | JSON]

Accuracy

Model/adapter	Mean win rate ↑ [sort]	MMLU - EM ↑ [sort]	BoolQ - EM ↑ [sort]	NarrativeQA - F1 ↑ [sort]	NaturalQuestions (closed-book) - F1 ↑ [sort]	NaturalQuestions (open-book) - F1 ↑ [sort]	QuAC - F1 ↑ [sort]	HellaSwag - EM ↑ [sort]	OpenbookQA - EM ↑ [sort]	TruthfulQA - EM ↑ [sort]
Cohere Command beta (52.4B)	**0.93**	0.452	0.856	**0.752**	0.372	0.76	0.432	0.811	0.582	0.269
text-davinci-002	0.93	0.568	0.877	0.727	0.383	0.713	0.445	0.815	0.594	**0.61**
text-davinci-	0.898	**0.569**	0.881	0.727	0.406	0.77	0.525	0.822	0.646	0.593

Disclaimer: metrics may not be up-to-date. Check https://crfm.stanford.edu/helm/latest for the latest.

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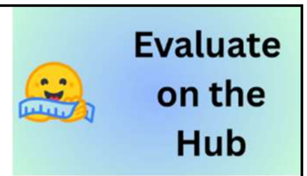
Huggingface Evaluator



Using the evaluator

The `Evaluator` classes allow to evaluate a triplet of model, dataset, and metric. The models wrapped in a pipeline, responsible for handling all preprocessing and post-processing and out-of-the-box, `Evaluator`s support transformers pipelines for the supported tasks, but custom pipelines can be passed, as showcased in the section Using the `evaluator` with custom pipelines.

Currently supported tasks are:

- `"text-classification"`: will use the TextClassificationEvaluator.
- `"token-classification"`: will use the TokenClassificationEvaluator.
- `"question-answering"`: will use the QuestionAnsweringEvaluator.
- `"image-classification"`: will use the ImageClassificationEvaluator.
- `"text-generation"`: will use the TextGenerationEvaluator.
- `"text2text-generation"`: will use the Text2TextGenerationEvaluator.
- `"summarization"`: will use the SummarizationEvaluator.
- `"translation"`: will use the TranslationEvaluator.
- `"automatic-speech-recognition"`: will use the AutomaticSpeechRecognitionEvaluator.
- `"audio-classification"`: will use the AudioClassificationEvaluator.

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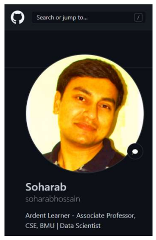

Thank You

Class material will be uploaded here:
https://github.com/soharabhossain/GenAI_n_LLMs

- Huggingface_Evaluator.txt
- LLM_Eval_Metrics.ipynb
- Semantic_Similarity.ipynb

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